

UNC303: COMPUTER AND COMMUNICATION NETWORKS

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3	0	0	3.0

Course Objective: To introduce basic concepts of Data communication with different models. Enumerate the physical layer, Data Link Layer, Network Layer, Transport Layer and Application Layer, explanation of the function(s) of each layer. Familiarization with cryptography and network security.

Overview of Data Communication and Networking: Data communications, Networks, The Internet, Protocols and standards, Layered tasks, OSI model, TCP /IP protocol Architecture and its addressing, Data Rate Limits, Circuit switching, Packet Switching, Message Switching.

Data link layer:

Types of errors, Detection, Error correction, Flow and error control, Stop and wait ARQ, go back n ARQ, Selective repeat ARQ, HDLC, Point to point protocol, PPP stack, ALOHA, CSMA, CSMA/CD, CSMA/CA, Reservation, Polling, Token Passing, FDMA, TDMA, CDMA. IEEE Standards: 802.3 to 802.6 and 802.11, FDDI, Bluetooth, SONET/SDH.

Network layer: Internetworks, Logical Addressing, Subnetting, Routing, ARP, IP, ICMP, IGMP, IPV6, Unicast routing, Unicast routing protocol, Multicast routing, Multicast routing protocols.

Transport layer: Process to process delivery, User datagram protocol (UDP), Transmission control protocol (TCP), Data traffic, Congestion, Congestion control, Quality of service, Techniques to improve QOS, Integrated services, Differentiated services.

Application layer: Client server model, Socket interface, Name space, Domain name space, Distribution of name space, DNS in the internet, Resolution, DNS messages, DDNS, Encapsulation, Electronic mail, File transfer, HTTP, World wide web (WWW), Network Management System, Cryptography, Network Security, Simple Network Management Protocol (SNMP), Simple Mail Transfer protocol (SMTP)

Familiarization with standard: Local and Metropolitan Area Networks-Overview and Architecture: IEEE 802-2014, Ethernet: IEEE 802.3, Token Bus: IEEE 802.4, Token Ring: IEEE 802.5, Wireless LAN (Wi-Fi): IEEE 802.11, Wireless Personal Area Networks (Bluetooth): IEEE 802.15.1.

Course Learning Objectives (CLO): The students will be able to:

1. Understand the layered architecture of Internet's reference models: OSI & TCP/IP and basis of physical layer and media.
2. Analyze the design issues, framing, error detection and correction, channel allocation techniques and link layer protocols.
3. Evaluate the network layer, internetworking, IP and IP addressing mechanism.
4. Apply various routing algorithms, elements of transport protocols, congestion control, and

QOS in networks.

5. Understand various communication applications like email, web browser, familiarization with cryptography and network security.

Text Books

1. Ferouzan, Behrouz A., Data Communications and Networking, TATA McGraw Hill 5th Edition, (2017)
2. Tanenbaum, Andrew S., Computer Networks, PHI 5th Edition, (2013).
3. D. Gross and C. Harris, Fundamentals of Queueing Theory, 3rd Edition, Wiley, 1998. (WSE Edition, 2004).

Reference Books

1. Stallings William, Data and Computer Communication, Pearson Education 10th Edition, (2017).
2. James F. Kurose, Computer networking: A top-down approach, Pearson Education 6th Edition, (2017).
3. Athanasios Papoulis, Probability Random Variables and Stochastic Processes, McGraw-Hill, 4th Edition, (2002).